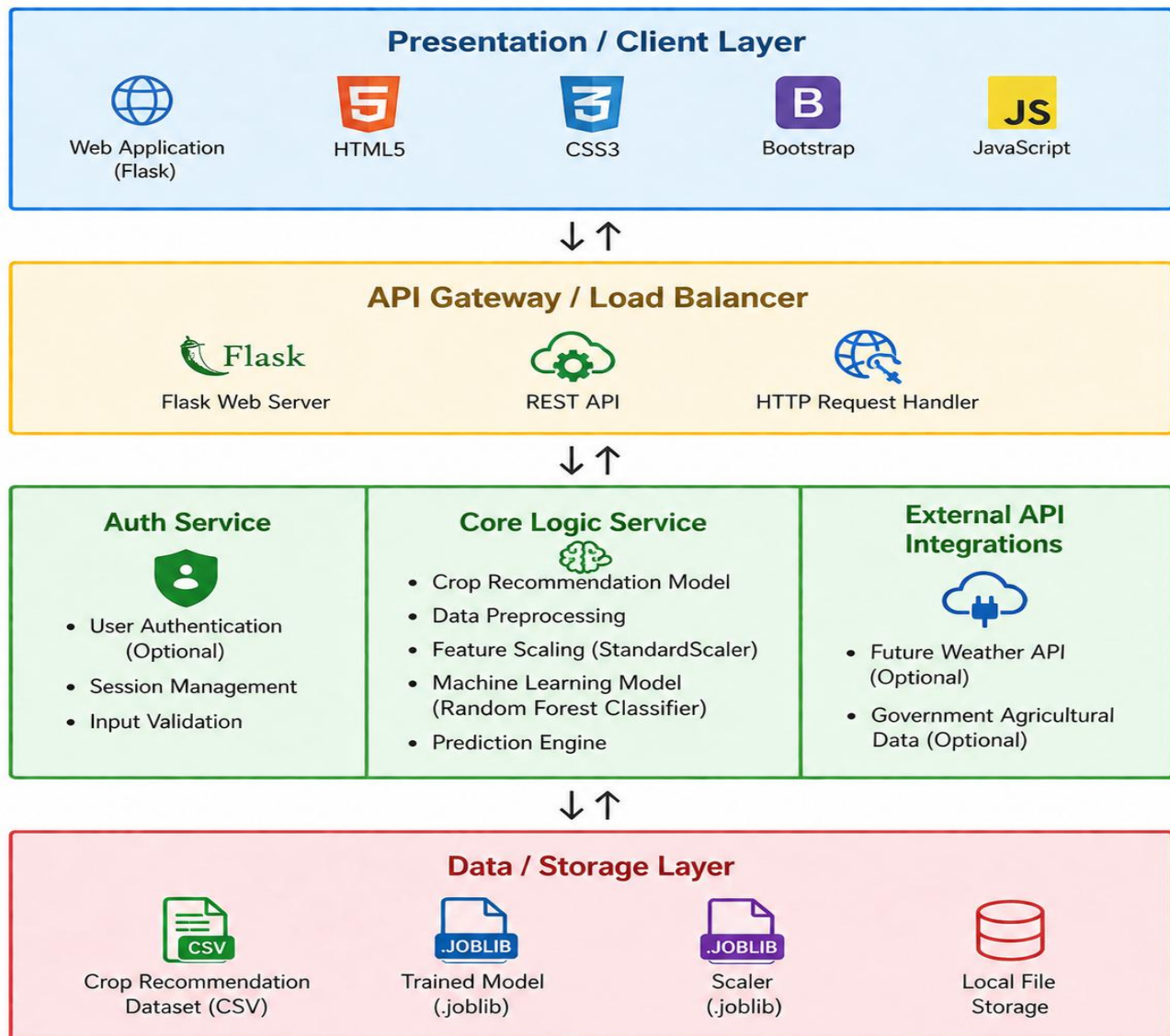


Solution Architecture

Date	1 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Solution Architecture Diagram:

The OptiCrop system adopts a layered architecture consisting of the Presentation Layer, Application Layer, Core Logic Layer, and Data Layer. It collects soil and environmental parameters, processes them using a trained machine learning model, and generates accurate crop recommendations through a Flask-based web application.



Component Description Table

Component Name	Description/Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly web interface where farmers enter soil and environmental parameters and view crop recommendations.	HTML5, CSS3, Bootstrap, JavaScript, Flask
API Gateway / Application Layer	Receives user requests, validates input data, and forwards them to the machine learning prediction service. It also returns prediction results to the user interface.	Flask, REST API, Python
Core Logic Service	Processes input data, performs feature scaling, executes the trained machine learning model, and predicts the most suitable crop based on soil and climate conditions.	Python, Scikit-learn, Random Forest Classifier, Joblib
Data / Storage Layer	Stores the crop recommendation dataset, trained machine learning model, scaler file, and other project resources required for prediction.	CSV Dataset, Joblib (.joblib), Local File Storage